Port Authority of NY and NJ Bus Terminal
Demand Forecasting and Capacity Planning 2026–2030

Executive Summary

The Port Authority Bus Terminal plays a critical role in regional transportation, handling millions of passengers annually. As the organization prepares infrastructure upgrades and temporary staging facility development, accurate forecasting of passenger demand is essential for ensuring operational efficiency, safety, and scalability. This report provides a comprehensive analytical solution to address five core planning questions posed by the Port Authority: terminal-level passenger volume projections through 2030, identification of the primary drivers of ridership, carrier-level demand trajectories, peak operational periods, and the terminal's recovery trajectory relative to pre-pandemic 2019 baselines. Using Prophet forecasting models, multiple linear regression, and interactive Power BI dashboards, the analysis provides predictive insights and planning recommendations.

The analysis reveals a consistent and sustained growth trajectory. Annual passenger volume is projected to increase from 5.6 million in 2026 to 6.85 million by 2030, representing 22% terminal-wide growth over the five-year forecast horizon. Average weekly peak volumes are expected to scale from approximately 108,000 to 132,000 passengers, with absolute peak weeks reaching as high as 142,500 passengers. This growth is not uniform across carriers: NJ Transit accounts for approximately 75% of all terminal volumes and drives the majority of infrastructure planning decisions.

The regression analysis identifies bus count, temperature, and snowfall as the three statistically significant drivers of weekly ridership, with the model explaining 92.5% of total ridership variance ($R^2 = 0.925$). Recovery analysis against the Fall 2019 baseline shows that the terminal reached approximately 84% of pre-pandemic volume by 2025, with full recovery projected by 2029 and a 108% surpass rate by 2030.

Data Overview:

The primary dataset used in this analysis consists of weekly ridership records sourced from the Port Authority Bus Terminal, covering the period from November 2020 through July 2025. After data cleaning — excluding holiday weeks, removing observations with missing passenger or bus count values, and filtering out weeks with zero bus operations — the final working dataset comprises 2,550 observations across twelve bus carriers, spanning 237 weeks.

Each observation represents a single carrier's operations for a given week and includes the following fields:

Column	Description	Role in Analysis
WeekStartDate	Start day of the week (Usually Monday)	Time index for all models
Bus Count	No. of weekly bus departures	Strongest ridership predictor
Passenger Count	No. of weekly passenger departures	Target (Y) variable
Carrier	Name of the Carrier	Segmentation Variable
Avgtemp, Avgsnow	Weekly averages for temperature; snow	External demand drivers
IsHolidayWeek, HolidayInWeek	Binary variable (1 = is a holiday week); name of the holiday	Seasonal flag adjustment

Holiday weeks were excluded from model training to prevent distortion of seasonal patterns, though holiday information was retained as an explicit input to the Prophet model via its dedicated holiday calendar. Two carriers — DeCamp and C&J Bus Lines — were excluded from the forecast due to insufficient data volume (fewer than 52 consecutive weeks of observation).

Forecasting Methodology

All passenger demand forecasts presented in this report were generated using Facebook's Prophet, an open-source time-series forecasting model developed for business forecasting applications. While the project evaluated alternative approaches including Exponential Smoothing (ETS), ARIMA, and SARIMAX during earlier phases of the analysis, Prophet was chosen based on its superior performance, strong fit with the data structure, and its ability to model the real-world dynamics of bus terminal demand.

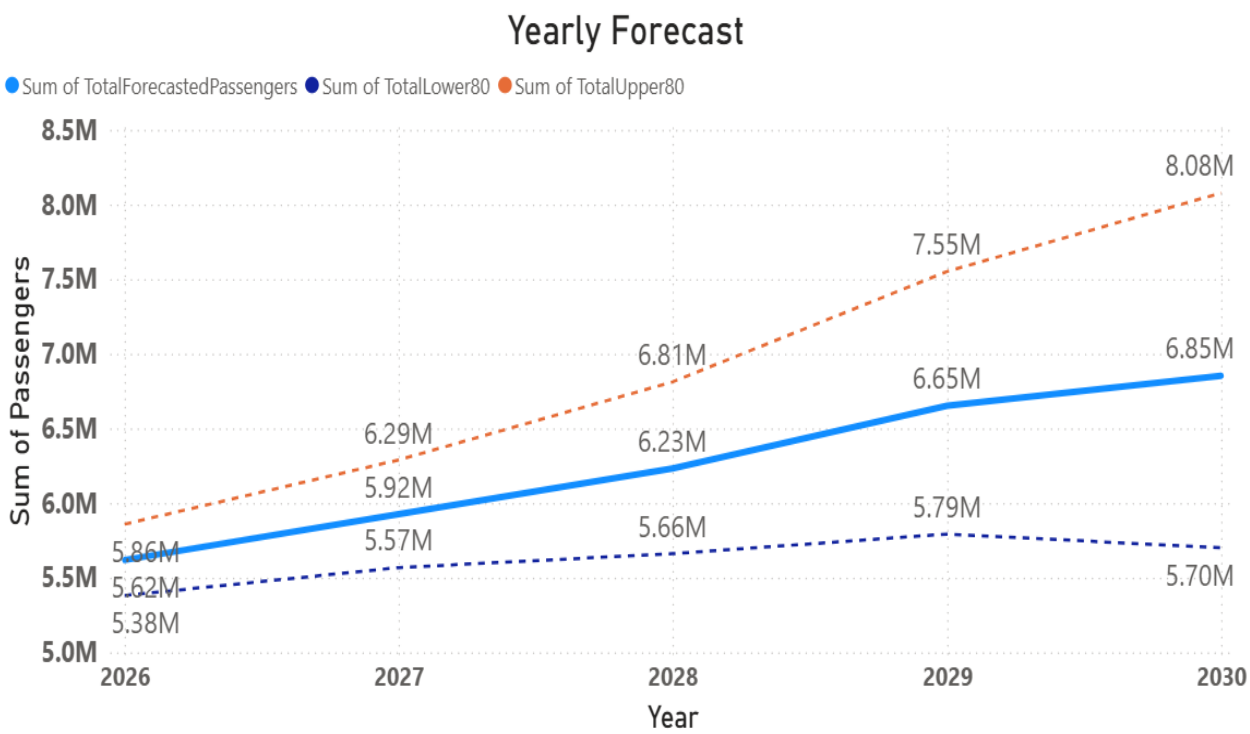
A key factor in this decision was performance on NJ Transit, which accounts for approximately 75% of total terminal volume. Prophet achieved an R^2 of 0.987 for this carrier, making it the most accurate model for the primary driver of overall demand. In addition, Prophet effectively captures important patterns in the data, including holiday effects, seasonal variation, and structural shifts such as COVID recovery. Its use of multiplicative seasonality ensures that peak demand scales appropriately as overall ridership grows, while automatic changepoint detection allows the model to adapt to trend changes without manual intervention.

Alternative models were not used for final forecasts due to key limitations. ETS cannot handle annual seasonality at a weekly level, making it unsuitable for this dataset. ARIMA struggled with seasonal structure and produced unrealistic projections for some carriers. While SARIMAX was appealing, Prophet performed consistently better over 7/10 forecasted carriers, and particularly NJ Transit.

Model performance was evaluated using an 80/20 train-test split, with final forecasts generated after refitting the full dataset. Mean Absolute Percentage Error (MAPE) was used as the primary accuracy metric. Six carriers achieved MAPE below 10%, indicating strong reliability for planning purposes. The remaining carriers showed higher volatility, and for these, forecast ranges (prediction intervals) should be emphasized over point estimates.

Q1: Terminal Passenger Forecast 2026 – 2030

The terminal-level forecast aggregates carrier-level projections to produce a composite view of total weekly and annual passenger demand through 2030. The central forecast line represents the best-estimate trajectory; the 80% prediction interval reflects the range within which actual outcomes are expected to fall. The difference between the best-estimate and the 80% prediction interval increases over time, reflecting increasing uncertainty as projections extend further into the future.



The forecasting analysis demonstrates a clear and sustained increase in passenger demand over the forecast horizon. Annual ridership is projected to grow from **5.6 million** passengers to **6.85** million by 2030, representing a robust **22%** total growth in terminal throughput. During this five-year window, average weekly peak volumes are expected to scale from approximately **108,000** to **132,000** passengers, with absolute peak weeks reaching as high as **142,500** travelers.

Year	Annual Total	Avg Weekly	Peak Week Est
2026	~ 5.6 M	~108,000	**~120,000**
2027	~ 5.9 M	~114,000	**~127,000**
2028	~ 6.2 M	~120,000	**~133,000**
2029	~ 6.7 M	~125,500	**~137,000**
2030	~ 6.85 M	~132,000	**~142,500**

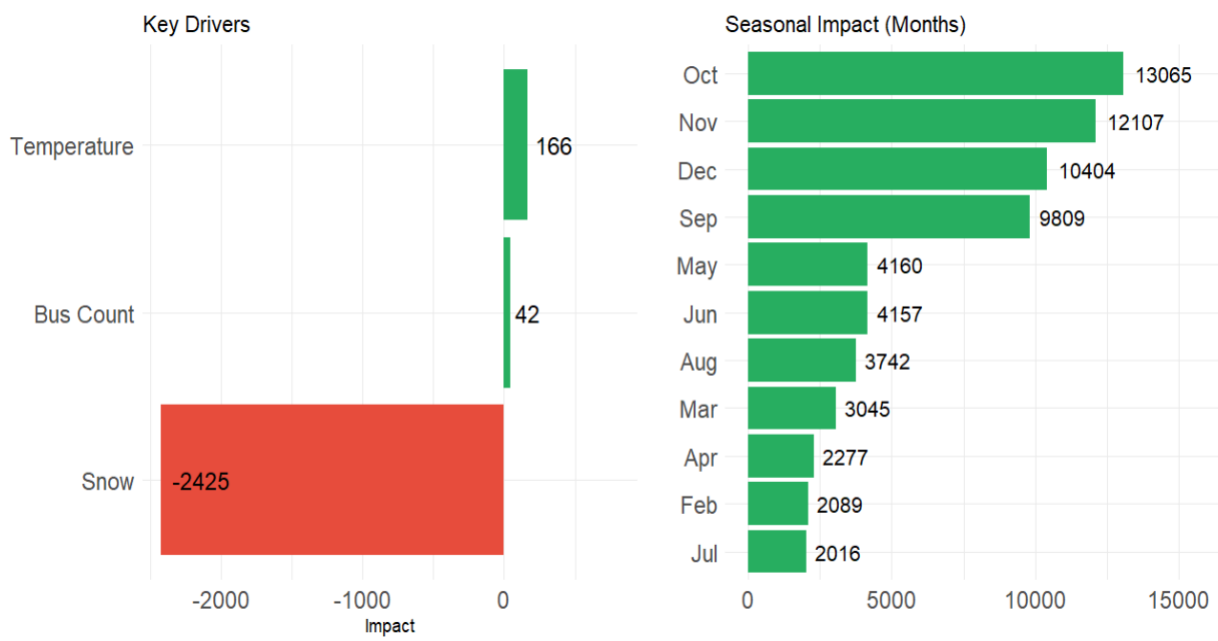
This consistent growth trajectory is driven by long-term behavioral and economic factors that necessitate the operational capacity to process roughly **3,400 additional passengers per day** by 2030 compared to 2026 levels. That is equivalent to adding a full rush-hour peak cycle of demand to the terminal's daily throughput, not a marginal adjustment, but a structural increase that must be built into infrastructure planning from the outset.

From a strategic business perspective, these findings carry significant implications for infrastructure development and operational readiness. The Port Authority must prioritize forward-looking, scalable capacity planning based on future peak demands rather than relying on historical averages or current utilization levels. Failure to align infrastructure with this projected growth could result in severe terminal congestion, systemic operational inefficiencies, and a marked decline in service quality.

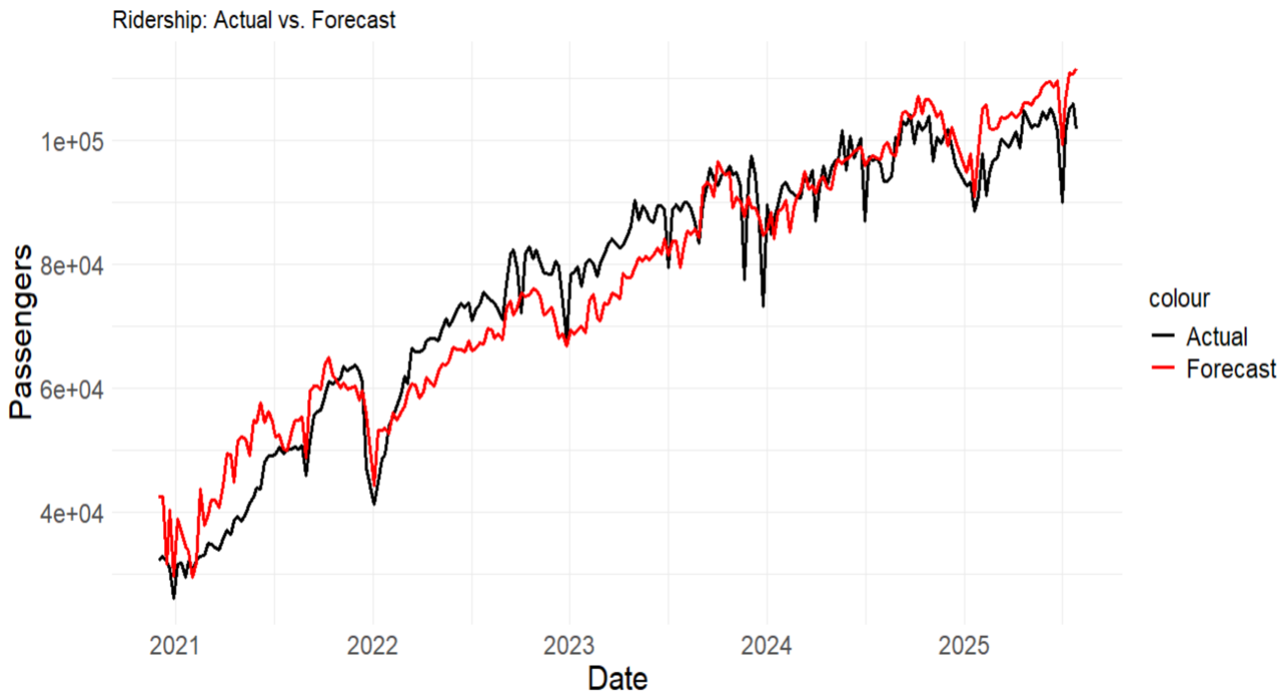
Q2: Key Drivers of Ridership

To understand the factors that systematically drive weekly passenger volume at the terminal level, a multiple linear regression model was developed using historical passenger counts as the dependent variable, with bus departures, temperature, and snowfall as predictors alongside year and monthly seasonal indicators. The full model specification is:

$$PassengerCount = -1.917\times10^{7} + 42.32\times BusCount + 166.0\times Temperature - 2{,}425\times Snowfall + 9{,}448\times Year + Monthly\ Seasonal\ Effects$$



The model achieves an R^2 of 0.925, indicating that the three predictor variables along with seasonal factors explain 92.5% of the total variance in weekly ridership. The MAPE on held-out data is 7.9%. These are strong results for a regression model applied to transportation demand data, and they support the use of the regression equation as a calibration tool for operational planning.



Bus Count

Bus count is the most operationally significant predictor, with a coefficient of 42.32 passengers per additional weekly bus departure. It is important to interpret this relationship carefully: bus count and passenger count are jointly determined. Carriers schedule buses in response to expected demand, and passengers ride because service is available. Rather than implying a simple causal direction, this coefficient functions as a planning ratio.

Temperature

Each 1°F increase in average weekly temperature is associated with approximately 166 additional terminal passengers. This coefficient reflects the well-documented relationship between warmer weather and increased travel activity, particularly for discretionary and leisure trips.

Snowfall

Snowfall has the largest coefficient magnitude in the model at negative 2,425 passengers per inch of weekly snowfall. A moderate winter storm delivering six inches of snow suppresses terminal ridership by approximately 14,550 passengers in a single week — representing roughly 11–14% of average weekly volume. This finding has direct implications for contingency planning: operational protocols for winter weather events should be established in advance of peak snow months.

Seasonal Impacts (Months)

The seasonal impact analysis confirms that October, November, and September show the highest month-level seasonality contributions — partially reflecting back-to-school and fall travel demand that co-occurs with moderate temperatures in the New York region.

Q3: Carrier Level Forecast

Carrier	2026	2028	2030	Trend
NJ Transit	4.8 M	5.3 M	5.8 M	↑ Dominant
Coach USA	307 K	336 K	366 K	↑ Growing
Peter Pan/Bonanza	111 K	161 K	211 K	↑ Strong
Academy	87K	105 K	123 K	↑ Steady
HCEE-Community	91 K	100 K	108 K	↑ Steady
TransBridge	34 K	34 K	34 K	→ Flat
Greyhound	53K	29 K	6 K	↓ Declining
Martz	37 K	36 K	36 K	→ Flat
Lakeland	74 K	75 K	76 K	↑ Steady

NJ Transit continues to dominate terminal demand, growing from 4.8 million riders in 2026 to 5.3 million in 2028 and 5.8 million in 2030. This sustained upward trajectory means NJ Transit alone accounts for the vast majority of terminal volume, making its service patterns and schedule changes the primary driver of overall terminal performance and congestion dynamics.

Among regional operators, several carriers show strong or steady growth. Peter Pan/Bonanza exhibits the most pronounced expansion, rising from 111K in 2026 to 161K in 2028 and 211K in 2030, more than doubling over the forecast horizon. Coach USA also shows consistent growth, increasing from 307K to 366K. Academy and HCEE Community demonstrate moderate but stable increases, while Lakeland edges upward gradually from 74K to 76K, indicating steady demand for retention.

In contrast, other carriers remain relatively flat or decline. TransBridge stays stable at 34K across all years, and Martz shows a slight taper from 37K to 36K before stabilizing. Greyhound experiences a significant decline, dropping sharply from 53K in 2026 to 29K in 2028 and down to just 6K by 2030. Overall, these trends highlight where capacity planning and operational coordination will need to be intensified for growing carriers, while also identifying services that may require restructuring or optimization due to stagnation or decline.

Passenger Per Bus: Operational Efficiency

Beyond aggregate demand forecasting, analyzing passengers per bus (PPB) provides a measure of how efficiently existing bus capacity is being utilized. A high PPB indicates that buses are running closer to full capacity; a low PPB indicates underutilized service, which represents a mismatch between supply and actual demand patterns.

The following table presents PPB by carrier from 2021 through 2025, with 56 seats used as a standardized capacity baseline for load factor calculation:

Carrier	2021	2022	2023	2024	2025	2025 Load Factor
Martz	38	40	40	41	41	**73%**
NJ Transit	18	26	31	34	35	**63%**
Lakeland	26	35	38	33	34	**61%**
Academy	23	31	32	30	32	**57%**
TransBridge	30	30	30	28	30	**54%**
Peter Pan/Bonanza	29	32	32	34	30	**54%**
Coach USA	24	31	33	29	29	**52%**
Trailways	25	22	25	25	21	**38%**
Greyhound	27	29	27	26	21	**38%**

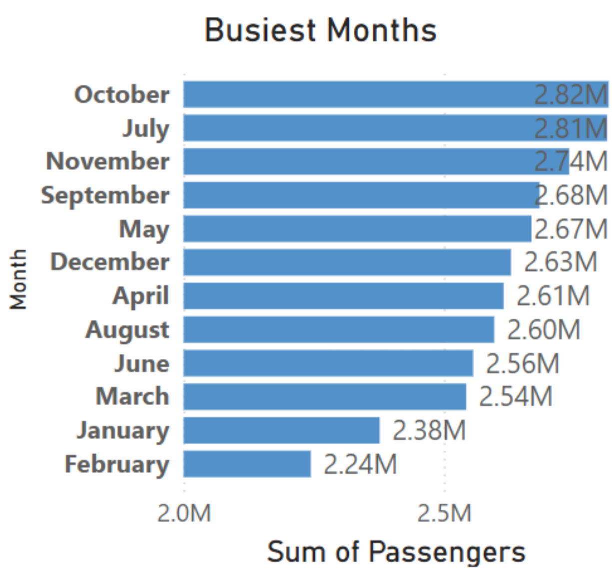
Martz achieves the highest utilization at 73%, while Greyhound and Trailways operate at only 38%, meaning nearly two-thirds of their average seat capacity goes unfilled each week. NJ Transit and Lakeland occupy the middle range at approximately 61–63%.

These disparities have direct operational implications. For high-load-factor carriers such as Martz, a continued increase in passenger volume will hit capacity constraints relatively quickly, requiring proactive service expansion planning. For low-load-factor carriers, the priority should be scheduling optimization — realigning departure times with actual demand patterns to reduce empty seat miles and improve cost efficiency. It is important to note that PPB varies significantly intra-week and intra-day, and these weekly averages mask the peak-hour utilization that drives real congestion at the terminal. Hourly or trip-level data would enable more precise gate-level planning.

Q4: Peak Year, Months, and Weeks

2030 is the peak year across the forecast horizon, with projected annual volume of 6.85 million passengers. At the monthly level, October carries the highest cumulative forecast across all five years at 2.82 million passengers, followed closely by July (2.81 million), November (2.74 million), and September (2.68 million). These four months consistently rank as the busiest across all forecast years and align with identifiable seasonal patterns: summer discretionary travel peaks in July, back-to-school and early fall travel drives September and October, and pre-Thanksgiving travel elevates November.

At the weekly level, weeks 38, 39, 43, and 44 carry the highest average single-week loads across the five-year forecast, each averaging over 129,000 passengers. These weeks correspond to mid-September through early November. Weeks 38 and 39 coincide with the start of the fall school semester and Labor Day travel recovery. Weeks 43 and 44 align with the approach to Halloween and pre-Thanksgiving weekend travel.



Busiest Weeks

WeekNum	Avg Passenger Forecast
44	129,524.54
43	129,224.92
38	129,093.16
39	129,011.20
49	128,083.58
45	126,981.44
42	126,889.38
40	126,757.82
50	126,391.57
48	126,056.91
37	125,721.76
41	125,509.72
46	124,023.93
47	123,663.20

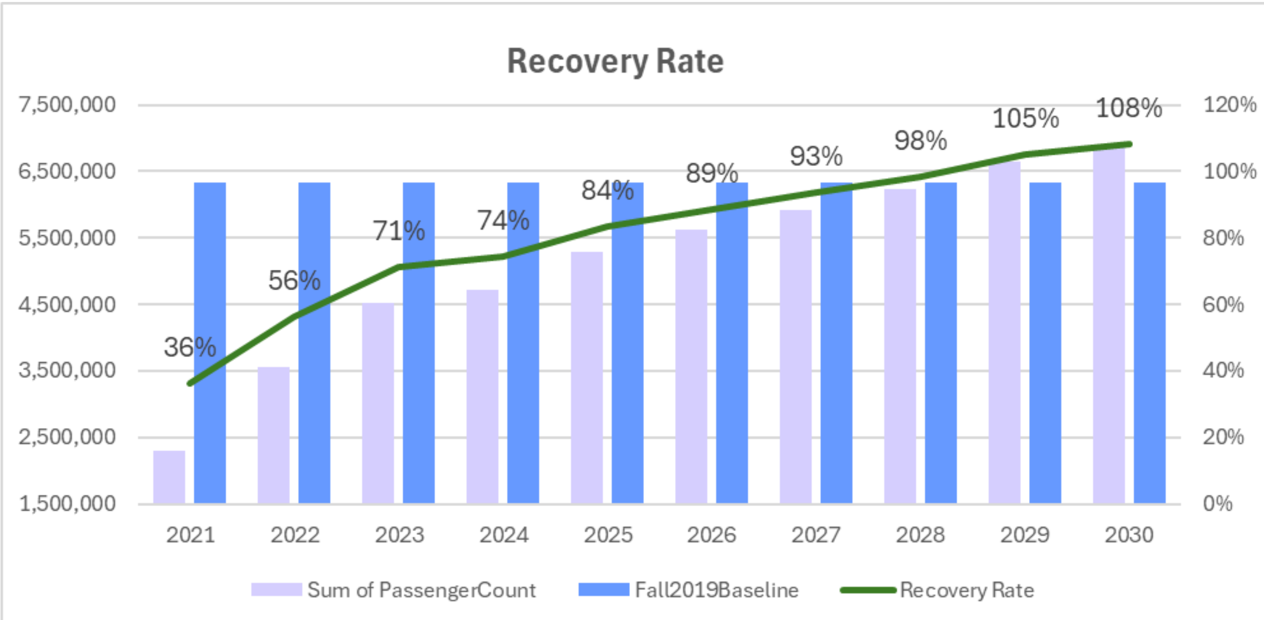
Operational Implications

These peak period findings translate directly into planning actions. Temporary staging facilities must be at full operational readiness by September 1st of each year to capture the September–November peak window. Staffing plans should reflect maximum headcount deployment during weeks 38–44. Gate assignments should prioritize NJ Transit and Peter Pan/Bonanza throughput during these weeks given their volume and growth trajectory respectively.

Conversely, January and February are the consistently lowest-volume months across all forecast years. These months represent the natural maintenance window: scheduled service reductions, facility renovations, gate closures, and equipment inspections should be concentrated in this period to minimize passenger impact.

Q5: Recovery vs. Fall 2019 Pre-COVID Baseline Comparison

The Fall 2019 comparison provides context for understanding how far terminal ridership has recovered relative to the last full pre-pandemic operating year. The recovery rate is calculated as the ratio of actual (and forecasted) annual passenger volume to the Fall 2019 equivalent baseline.



In 2021, ridership was heavily suppressed at approximately 2.3 million passengers, representing just 36% of baseline levels due to the impact of COVID-19. Recovery began in 2022, with volume rising to 3.56 million (56%), marking the early stages of demand returning. By 2023, ridership reached 4.5 million (71%), indicating a faster, accelerating recovery phase. In 2024, the passenger volumes reached 4.7 million at 74% recovery. By the end of July 2025, the sum of passenger volume reached ~2.98 million and is projected to increase to 5.1 million at 84% recovery by the end of 2025.

From 2026 onward, the data reflects all forecasted growth. Passenger volume is expected to steadily increase from 5.6 million (89%) in 2026 to 6.2 million (98%) by 2028, nearly reaching the 2019 baseline. Full recovery is projected in 2029, when ridership surpasses pre-pandemic levels at 6.65 million (105%). By 2030, the terminal enters a full expansion phase, with demand reaching 6.85 million passengers, or 108% of the baseline.

Strategic Recommendations

Based on forecasting, carrier-level analysis, and recovery trends, several strategic actions are recommended to ensure the Port Authority Bus Terminal remains operationally efficient and prepared for future growth.

Design Temporary Staging Facilities for Future Peak Demand

The temporary staging facility should be designed for approximately 171,000 passengers per week by 2030, based on projected peak demand and a 20% operational buffer for uncertainty and surge events. Designing only for near-term demand would likely create congestion and operational bottlenecks by 2028. While modest over-sizing increases upfront costs, under-sizing risks chronic overcrowding, service disruptions, and safety concerns at one of the nation's busiest transit hubs.

Expand the Regression Model with Economic Variables

The regression model explains 92.5% of ridership variance using bus count, temperature, and snowfall. The remaining variance likely reflects broader economic and behavioral factors such as employment trends, fuel prices, remote work, and major events. Incorporating these variables into future models could improve forecast accuracy and provide earlier insight into demand shifts.

Monitor NJ Transit Closely and Establish Data Sharing

NJ Transit accounts for approximately 75% of total terminal volume, making it the primary driver of staging and capacity planning decisions. Its projected growth from 4.37 million passengers in 2026 to 5.13 million by 2030 means that service disruptions or schedule changes could significantly impact terminal operations. To improve forecast accuracy and operational preparedness, the Port Authority should establish a formal data-sharing partnership with NJ Transit to integrate future service and schedule information into forecasting models.

Optimize Bus Scheduling and Service Frequency

The regression analysis identified Bus Count as the strongest controllable factor affecting passenger demand. The passenger-per-bus ratio can be used as a planning ratio: dividing the passenger forecast for a given week by the expected PPB for that carrier yields the approximate number of bus trips required to maintain current load efficiency. The Port Authority should improve scheduling practices by increasing bus frequency during peak periods and improving coordination among carriers. Better scheduling can reduce congestion, improve passenger flow, and maximize terminal utilization.

Implement Seasonal Staffing Strategies

Passenger traffic varies throughout the year, with March, April, August, and September showing the highest demand levels. The Port Authority should adopt a flexible staffing strategy that increases workforce allocation during peak periods and optimizes staffing during lower-demand months. This approach can improve customer service, crowd management, and operational efficiency.

Establish a Continuous Forecasting and Monitoring Process

Passenger demand patterns continue to evolve due to changing travel behavior, economic conditions, and external disruptions. Forecasting should therefore become an ongoing business process rather than a one-time analysis. The Port Authority should establish a continuous forecasting framework with quarterly or monthly model updates using newly available operational data. Automating forecasting pipelines and reporting dashboards would support faster, data-driven decision-making and improve responsiveness to changing demand patterns.

Enhance the Use of Business Intelligence and Analytics Tools

The Power BI dashboard developed in this project demonstrates the value of analytics tools in supporting operational and strategic decisions. The Port Authority should continue expanding the use of business intelligence platforms to improve visibility into passenger trends, operational performance, and congestion patterns. Interactive dashboards and real-time reporting tools can improve communication across departments and strengthen the organization’s data-driven decision-making culture.

Conclusion

This project delivers a rigorous, data-driven framework for anticipating and managing passenger demand at one of the nation's busiest and most operationally complex transit hubs. The Port Authority Bus Terminal faces increasing pressure to manage rising passenger demand while maintaining service quality, operational reliability, and infrastructure efficiency. By combining Prophet forecasting, regression analysis, and Power BI dashboards, the analysis produces actionable projections across five planning dimensions terminal volume, ridership drivers, carrier trajectories, peak periods, and COVID recovery with measurable accuracy. The results are not theoretical: they translate directly into gate assignments, staffing calendars, staging facility specifications, and carrier coordination priorities. The dashboard developed as part of this project enables management teams to monitor trends, evaluate operational performance, and make faster, data-driven decisions.

This supports a transition from reactive decision-making toward a more proactive and predictive operational model.

The overarching implication is clear since the Port Authority Bus Terminal is not approaching a period of stabilization, but structural expansion. With annual ridership projected to reach 6.85 million by 2030 and peak weeks exceeding 142,000 passengers, the window for proactive planning is now. Organizations that embed continuous forecasting into operational decision-making will be positioned to absorb this growth seamlessly; those that do not risk congestion, service degradation, and infrastructure failure now of peak demand.